Evidence Before Sizing:
An Operational Audit Protocol for XRP-Adjacent Institutional Infrastructure

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Abstract

This paper develops an operational audit protocol for determining when XRP-adjacent institutional infrastructure becomes eligible for downstream filtered liquidity sizing. Earlier papers in this series distinguished visible usage from retained demand, infrastructure expansion from settlement closure, retained demand from required liquidity density, and institutional stack expansion from asset-specific amplification. The present paper asks a procedural question: before an XRP-adjacent layer can be admitted into a sizing exercise, what evidence must be present, in what order, and under what conditions should the layer be closed, deferred, or elevated?

The paper does not provide a price forecast, investment thesis, or claim of XRP indispensability. Its contribution is procedural rather than theoretical. Papers 1 through 4 established what questions must be asked. This paper specifies the order in which those questions should be answered, the evidence standard each question requires, and the conditions under which an XRP-adjacent layer may become an admissible input for filtered liquidity sizing.

The framework combines an evidence gradient, five eligibility gates, institutional friction scoring, removal-sensitivity analysis, proxy indicators for hard-to-observe layers, and exit criteria for bypass or compression outcomes. The five eligibility gates are retained demand, selection, institutional friction, liquidity-density burden, and removal sensitivity. These gates require that XRP be not merely adjacent, available, or technically supported, but retained, selected, institutionally admissible, liquidity-relevant, and costly to remove.

A central contribution of the paper is the Institutional Friction Gate, which distinguishes technical possibility from institutional adoption. It evaluates whether XRP can be retained, governed, risk-managed, accounted for, and operationally relied upon under the balance-sheet, accounting, regulatory, governance, custody, compliance, incentive, and installed-base constraints that shape institutional behavior.

The framework is applied to selected XRP-adjacent layers, including Ripple Payments, RLUSD, XRPL activity, XRPL AMM / DEX liquidity, Ripple Custody, Ripple Prime / Hidden Road, Ripple Treasury, tokenization and RWA settlement, collateral and margin infrastructure, derivatives and hedging, market-making inventory, stress-time liquidity, cross-ledger settlement, and agent commerce. Current public evidence supports strong XRP-adjacent domain expansion, but does not establish stack-wide XRP-specific amplification.

The paper's substantive conclusion is that the decisive future evidence zones are narrower than the overall XRP-adjacent ecosystem. The most important layers for future sizing eligibility are market-making inventory, collateral and margin, Ripple Prime / Hidden Road, stress-time liquidity, institutional XRPL liquidity, and cross-ledger settlement. RLUSD and tokenization / RWA remain strategically important but

are not XRP-specific sizing inputs unless XRP-mediated liquidity, retained balances, institutional admissibility, or removal-sensitive workflows emerge.

The paper treats public sizing literature as useful but downstream. Sizing is not rejected; sizing is gated. A model that sizes every adjacent narrative becomes noisy, while a model that sizes only gate-passing layers becomes sharper. The contribution of this paper is therefore methodological and applied: it defines when XRP-adjacent evidence becomes admissible for filtered liquidity sizing, and when it should remain outside the model.

1. Introduction: Evidence Before Sizing

Institutional digital-asset infrastructure is expanding across payment networks, stablecoins, custody platforms, prime-brokerage arrangements, treasury tools, tokenization initiatives, collateral systems, automated market-making infrastructure, and cross-ledger settlement designs. This expansion has made the evaluation of institutionally connected digital assets both more important and more difficult. As more institutional layers become adjacent to a digital asset, the temptation grows to convert infrastructure growth into asset-specific demand claims.

The preceding papers in this series were designed to slow that inference. Paper 1 distinguished visible usage from retained demand. Paper 2 distinguished infrastructure expansion from settlement closure. Paper 3 distinguished conditional liquidity sizing from price narrative. Paper 4 distinguished stack expansion from asset-specific amplification through the categories of compression, bypass, and amplification.

Within the retained-demand audit series, this paper functions as the fifth layer: an operational eligibility layer between stack-outcome analysis and downstream filtered sizing. It does not add a new asset-demand category; it defines when evidence is sufficient for an XRP-adjacent layer to move from narrative proximity or infrastructure readiness into sizing admissibility.

The present paper asks a prior applied question: before XRP-adjacent institutional infrastructure can be sized, what evidence must be present for a layer to become an admissible input? This is the problem of evidence before sizing.

The contribution of this paper is procedural rather than theoretical. Papers 1 through 4 established what questions must be asked. This paper specifies the order in which those questions must be answered, the evidence standard each question requires, and the conditions under which a layer should be closed, deferred, or elevated to sizing eligibility. Without this sequencing, the prior frameworks remain structurally correct but operationally incomplete.

This paper treats filtered liquidity sizing as analytically powerful but input-sensitive. Its purpose is therefore not to reject sizing, but to define the evidentiary and institutional conditions under which a layer may be admitted into sizing without converting narrative proximity into numerical inference. Put differently, the purpose is not to reject sizing. The purpose is to determine when sizing is earned.

The paper therefore does not provide a price forecast, investment thesis, or claim of XRP indispensability. It defines an operational audit protocol for determining when XRP-adjacent institutional layers become eligible for downstream filtered liquidity sizing.

The protocol has five core components. First, it introduces an evidence gradient. Second, it defines five eligibility gates: retained demand, selection, institutional friction, liquidity-density burden, and removal sensitivity. Third, it introduces an institutional friction score. Fourth, it treats removal sensitivity as the final gate. Fifth, the protocol includes positive triggers, negative triggers, proxy indicators, and exit criteria. A serious audit protocol must be able to close gates as well as open them.

The framework is applied to selected XRP-adjacent institutional layers, including Ripple Payments, RLUSD, XRPL base activity, XRPL AMM and DEX liquidity, Ripple Custody, Ripple Prime / Hidden Road, Ripple Treasury, tokenization and RWA settlement, collateral and margin infrastructure, derivatives and hedging, market-making inventory, stress-time liquidity, cross-ledger settlement, and agent commerce or machine payments.

The paper's central conclusion is limited but substantive. Current public evidence supports strong XRP-adjacent domain expansion, but it does not establish stack-wide XRP-specific amplification. However, the universe of relevant evidence is now narrower and more operationally defined. The decisive question is whether a limited set of layers - market-making inventory, collateral and margin, prime brokerage, stress-time liquidity, institutional XRPL liquidity, and cross-ledger settlement - can produce evidence of retained demand, institutional admissibility, liquidity-density burden, and removal-sensitive dependence.

This formulation is deliberately neither promotional nor dismissive. It does not infer XRP demand from ecosystem growth. It also does not dismiss XRP relevance simply because public evidence remains incomplete. Instead, it asks where the decisive evidence would have to appear, what form that evidence would take, and which layers should remain outside filtered liquidity sizing until those conditions are satisfied.

Throughout this paper, XRP-adjacent infrastructure is therefore not treated as XRP retained demand. Adjacency, technical support, product integration, or ecosystem expansion may justify monitoring, but they do not justify filtered sizing unless the relevant layer passes the retained-demand, selection, institutional-friction, liquidity-density, and removal-sensitivity gates.

The remainder of the paper proceeds as follows. Section 2 explains how the paper moves from stack outcomes to sizing eligibility. Section 3 introduces the evidence gradient. Section 4 defines the five eligibility gates and the institutional friction score. Section 5 maps the XRP-adjacent institutional layers considered in the paper. Section 6 applies the evidence-gated classification framework to those layers. Section 7 develops removal sensitivity as the final gate. Section 8 identifies high-priority evidence zones, proxy indicators, and exit criteria. Section 9 clarifies the relationship between this framework, filtered liquidity sizing, and public sizing literature. Section 10 concludes.

2. From Stack Outcomes to Sizing Eligibility

Paper 4 established that institutional stack expansion can produce different asset-specific outcomes. A growing stack may compress the need for asset-specific retained liquidity, bypass the asset while expanding adjacent infrastructure, or amplify the need to retain, mobilize, post, buffer, or maintain inventory in the asset itself. This distinction is essential, but it does not by itself determine whether a layer is ready for liquidity-density sizing.

A stack outcome classification answers one question: what direction does a given institutional layer appear to move asset-specific retained demand? Sizing eligibility answers a different question: has that layer produced enough evidence to justify translating the retained-demand claim into a conditional liquidity-density sizing exercise?

These questions are related, but they are not identical. A layer may be classified as potential amplification without being sizing-eligible. A layer may be structurally important but still lack evidence of retained inventory, actual selection, institutional admissibility, liquidity-density burden, or removal-sensitive dependence. Conversely, a layer may not be the most visible part of the ecosystem, yet may become sizing-relevant if it produces direct evidence of inventory, collateral, market-making, stress-time liquidity, or removal degradation.

Paper 5 therefore functions as a transition layer between Paper 4 and Paper 3. Paper 4 identifies possible stack outcomes. Paper 3 explains how retained demand, if operationally meaningful, becomes a quantity

problem involving usable float, available depth, turnover, positioning, slippage tolerance, and bounded market impact. Paper 5 asks which layers have enough evidence to move from stack-outcome classification into that quantity problem.

Stage	Question	Output
Paper 1	Is the asset merely used, or must it be retained?	Retained-demand framework
Paper 2	Does infrastructure expansion leave economically material residual overflow?	Closure / overflow / selection framework
Paper 3	If retained demand exists, what liquidity-density conditions are required?	Conditional sizing framework
Paper 4	Does the stack compress, bypass, or amplify asset-specific retained demand?	Stack-outcome framework
Paper 5	Which layers are admissible inputs for filtered liquidity sizing?	Operational audit protocol

The key contribution of Paper 5 is therefore not another category of stack outcome. It does not add a fourth outcome beyond compression, bypass, and amplification. Instead, it defines the evidentiary and institutional threshold required before a layer can move from outcome classification to filtered sizing.

Compression generally weakens or delays sizing eligibility. If a layer reduces the need for XRP-specific retained balances through netting, internalization, treasury optimization, stablecoin settlement, custody aggregation, or collateral efficiency, then the liquidity-density burden may be lower than a surface reading of gross activity would imply.

Bypass normally blocks sizing eligibility. If a layer expands while XRP remains absent, optional, replaceable, or weakly retained, then XRP-specific liquidity sizing is not warranted. The layer may be institutionally important, but the asset has not been selected in a way that justifies XRP-specific sizing.

Amplification may open the door to sizing eligibility, but only if evidence is strong enough. A layer that plausibly increases XRP-specific retained demand still must show retained balances, realized selection, institutional admissibility, liquidity-density burden, and removal-sensitive dependence before filtered sizing becomes analytically appropriate.

A sizing-eligibility framework prevents two opposite errors. The first is premature sizing, in which an analyst observes an XRP-adjacent development and immediately translates it into settlement-flow assumptions or implied valuation ranges. The second is premature dismissal, in which incomplete public evidence is taken to prove irrelevance even though the relevant retained-demand evidence may be internal, emerging, stress-contingent, or difficult to observe.

Only then is filtered liquidity sizing earned.

3. Evidence Gradient

A sizing-eligibility framework requires a way to evaluate incomplete evidence without collapsing analysis into either overclaim or dismissal. Public information about institutional digital-asset infrastructure is often partial. Some elements are visible: product announcements, stablecoin issuance, custody support, exchange access, tokenization initiatives, AMM deployment, or prime-brokerage expansion. Other elements are much less visible: retained balances, liquidity-provider inventory, market-maker positioning, collateral posting, internal routing, netting structures, stress-time liquidity behavior, and removal-sensitive operational dependence.

For that reason, evidence relevant to XRP-specific retained demand should not be treated as binary. A layer is rarely either fully proved or fully irrelevant. It may be adjacent but unselected. It may be infrastructure-ready but not yet used. It may show signs of selection but no retained balances. It may be technically usable but institutionally constrained. It may show retained demand but not yet a meaningful liquidity-density burden. It may show liquidity burden but not removal-sensitive dependence.

This paper therefore introduces an evidence gradient. The gradient is designed to classify how far a layer has progressed toward sizing eligibility. It does not measure price significance. It measures evidentiary maturity.

Level	Evidence state	Meaning
Level 0	Narrative adjacency only	The layer is associated with XRP, Ripple, XRPL, RLUSD, or institutional digital-asset narratives, but no operational XRP role is shown.
Level 1	Infrastructure ready / selection pending	Infrastructure exists that could support XRP use, but XRP selection is not yet shown.
Level 2	Selection indicated / institutional admissibility unproven	XRP appears selected or usable in a workflow, but institutional admissibility, retained balances, or sustained use remain unproven.
Level 3	Retained demand / institutional use evidence emerging	Evidence suggests XRP is held, positioned, posted, buffered, or inventoried for operational reasons.
Level 4	Liquidity-density burden visible	XRP-specific retained demand appears to impose meaningful burden on usable float, depth, turnover, positioning, or slippage tolerance.
Level 5	Removal-sensitive dependence established	Removing XRP would materially worsen cost, timing, slippage, liquidity access, collateral mobility, capital efficiency, stress-time execution, or resilience.

The purpose of this gradient is not to make weak evidence appear stronger. It is to prevent weak evidence from being misclassified and to prevent partially meaningful evidence from being ignored. A Level 1 layer is not a Level 5 layer. But it is also not analytically identical to pure narrative speculation. Similarly, a Level 2 layer may show signs of selection but still fail institutional admissibility or retained-demand tests.

The threshold between Level 2 and Level 3 is the presence of workflow-linked holding behavior: evidence that the asset is maintained in retained form as part of an institutional process, rather than merely appearing in transactional flow or being technically available.

Level 3 is significant because it moves the analysis from visibility to balance-sheet or operational positioning. However, Level 3 does not automatically imply sizing eligibility. The retained balances may be small, dispersed, high-turnover, institutionally constrained, or operationally limited. To reach sizing relevance, the analysis must still determine whether those retained balances create a meaningful liquidity-density burden and whether XRP is costly to remove.

Level 4 is where the required-liquidity-density framework becomes directly relevant. It is stronger than retained demand alone because it shows that the retained demand is not merely present, but materially burdensome. Level 5 is the strongest evidence level because it links retained demand to comparative operational necessity.

Evidence level	Sizing implication
Level 0	Not sizing-eligible
Level 1	Not sizing-eligible; infrastructure-ready monitoring only
Level 2	Not yet sizing-eligible; selection or technical integration indicated, but institutional admissibility and retained demand unproven
Level 3	Potential sizing candidate if retained balances are meaningful and institutionally admissible
Level 4	Sizing-relevant if liquidity-density burden is sufficiently specified
Level 5	Strongest sizing eligibility; removal-sensitive dependence established

The evidence gradient also helps assign classification confidence. A layer can be classified directionally, but the confidence of that classification depends on evidence level, observability, and the complexity of possible stack outcomes.

The key point is simple: XRP-adjacent infrastructure does not become sizing-eligible because it is visible, important, or growing. It becomes sizing-eligible only as evidence moves up the gradient from adjacency

to selection, from selection to institutional admissibility and retained demand, from retained demand to liquidity-density burden, and from liquidity burden to removal-sensitive dependence.

4. The Five Eligibility Gates

The evidence gradient identifies how far a layer has progressed toward sizing eligibility. This section defines the gates that determine whether an XRP-adjacent institutional layer can move from evidence classification into downstream filtered liquidity sizing.

Earlier versions of the framework used four gates: retained demand, selection, liquidity-density burden, and removal sensitivity. Those gates remain necessary, but they are not sufficient. Between technical selection and measurable liquidity burden lies a critical institutional question: can the asset be admitted, governed, accounted for, risk-managed, and held by real institutional actors? This paper therefore introduces a fifth gate: the Institutional Friction Gate.

The five eligibility gates are: Retained Demand Gate, Selection Gate, Institutional Friction Gate, Liquidity-Density Burden Gate, and Removal-Sensitivity Gate. Together, these gates define the evidentiary path from XRP adjacency to filtered liquidity sizing.

A layer does not become sizing-eligible because it is large, visible, or strategically interesting. It becomes sizing-eligible only if the evidence shows that XRP is retained, selected, institutionally admissible, liquidity-burdensome, and removal-sensitive in the relevant workflow.

4.1 Gate 1: Retained Demand Gate

The retained-demand gate asks whether XRP is held, positioned, posted, buffered, or inventoried in operationally meaningful form. This gate is the first threshold because the series begins from the distinction between visible usage and retained demand.

The gate requires evidence that XRP is held because the workflow needs it. Relevant evidence includes XRP retained by liquidity providers or market makers, held as settlement inventory or corridor liquidity, posted as collateral or margin, maintained as liquidity buffer or waiting liquidity, or retained inside treasury, prime-brokerage, or institutional liquidity workflows. Without retained demand, there is no meaningful sizing problem.

4.2 Gate 2: Selection Gate

The selection gate asks whether XRP is actually selected within the relevant workflow. This gate is necessary because residual overflow, infrastructure readiness, or technical possibility does not imply asset selection.

The selection gate requires evidence that XRP is chosen for a role: settlement routing, bridge liquidity, liquidity provision between assets, collateral or margin workflows, RLUSD-related liquidity paths, tokenized-asset settlement, cross-ledger settlement, or stress-time liquidity conditions. If stablecoins, tokenized deposits, fiat rails, internal treasury systems, bridges, or orchestration layers perform the function without XRP, then XRP-specific sizing is not warranted.

4.3 Gate 3: Institutional Friction Gate

The institutional friction gate asks whether XRP can be practically admitted into institutional balance sheets, risk systems, governance structures, compliance processes, and operational workflows. This gate is necessary because technical selection is not institutional adoption.

An asset may be fast, liquid, inexpensive, or technically compatible, yet still fail institutional use because it cannot be comfortably held, accounted for, governed, approved, risk-managed, or justified inside

regulated financial entities. The institutional friction gate therefore examines the gap between 'XRP can be used' and 'XRP can be institutionally retained, governed, and relied upon.'

Friction Type	Audit Question
Regulatory treatment	Can regulated institutions hold or use XRP within applicable rules?
Accounting treatment	Can XRP be carried, valued, and reported in a manageable way?
Balance-sheet treatment	Does XRP create unacceptable balance-sheet volatility or capital burden?
Governance approval	Can internal committees approve XRP use or retention?
Risk limits	Can XRP fit within volatility, liquidity, concentration, VaR, or exposure limits?
Custody and control	Can XRP be securely custodied, controlled, audited, and transferred?
Compliance operations	Can AML, sanctions, counterparty, and transaction-monitoring requirements be met?
Incentive alignment	Who has an economic reason to retain XRP, rather than merely route through it?
Installed base	Are there enough counterparties, venues, liquidity providers, and institutional users to support the workflow?
Substitution pressure	Are stablecoins, tokenized deposits, fiat rails, or internal ledgers easier to approve?

4.3A Institutional Friction Score

Unlike the selection constraints identified in Paper 2, which analyzed structural barriers to asset selection at the stack level, the Institutional Friction Score evaluates asset-specific admissibility within actual institutional workflows: whether the asset can be retained, governed, risk-managed, and operationally relied upon under the specific balance-sheet, accounting, and regulatory conditions that govern institutional behavior.

The institutional friction gate should not remain a purely qualitative checklist. In practice, institutional admissibility exists on a spectrum. An asset may be technically supported but not governance-approved. It may be custody-ready but balance-sheet constrained. It may be legally permissible but unattractive under risk, accounting, or capital treatment. It may be tradable but not retainable.

For that reason, this paper introduces a qualitative Institutional Friction Score. The score is not intended to produce numerical precision. It is intended to prevent the analysis from treating all institutionally adjacent layers as equally admissible.

Friction Score	Institutional State	Meaning
F0	Not institutionally admissible	The asset or workflow cannot be practically used by regulated institutions under current constraints.
F1	Technically supported, institutionally unresolved	Infrastructure exists, but regulatory, accounting, governance, or risk treatment remains unclear.
F2	Operationally available, balance-sheet constrained	Custody, compliance, or trading access may exist, but retention is limited by balance-sheet, capital, volatility, or risk constraints.
F3	Institutionally usable under limited conditions	The asset can be used or retained in defined workflows, with constraints on size, duration, counterparties, or purpose.
F4	Broadly admissible with manageable constraints	The asset can be integrated into institutional workflows with manageable risk, accounting, custody, and governance treatment.
F5	Fully integrated institutional workflow asset	The asset is routinely retained, governed, risk-managed, and operationally relied upon within institutional workflows.

The score should be applied conservatively. Public evidence may support a layer's technical availability, but not its institutional admissibility. Custody support may move a layer from F0 to F1 or F2, but it does not automatically move it to F4 or F5. Actual institutional retention, collateral use, margin acceptance, risk-limit integration, or treasury usage would be required for higher scores.

In this sense, the Institutional Friction Score acts as a bridge between technical possibility and institutional retained demand. It asks not only whether XRP can be used, but whether it can be approved, held, governed, risk-managed, and relied upon by the institutions whose behavior would matter for sizing.

4.4 Gate 4: Liquidity-Density Burden Gate

The liquidity-density burden gate asks whether XRP-specific retained demand creates meaningful pressure on usable float, available depth, turnover, positioning, execution windows, or slippage tolerance. This gate connects the present paper to the required-liquidity-density framework.

Retained demand does not automatically imply a large or meaningful sizing burden. XRP may be selected and retained in small amounts, in highly distributed ways, with fast turnover, wide slippage tolerance, or minimal operational consequence. The gate requires evidence that retained XRP demand imposes a non-trivial liquidity burden.

4.5 Gate 5: Removal-Sensitivity Gate

The removal-sensitivity gate asks whether removing XRP from the relevant workflow would materially worsen the workflow relative to actual alternatives. This is the final and strongest gate. It does not ask whether XRP is present. It asks whether XRP matters.

A layer may show retained demand, selection, institutional admissibility, and some liquidity burden, but if XRP can be removed without meaningful deterioration, the stronger sizing claim remains weak. Conversely, even if XRP is not dominant across an entire system, it may become structurally meaningful if removal produces measurable degradation under identifiable conditions.

The final question is not whether XRP is present. It is whether XRP is institutionally admissible, operationally retained, liquidity-relevant, and costly to remove.

4.6 Gate Status and Sizing Eligibility

Gate Status	Meaning
Closed	The gate is not passed on current evidence.
Partially open	Some evidence exists, but not enough for stronger classification.
Open	Public or credible evidence meaningfully supports the gate.
Hard-to-observe	The gate may be relevant, but key evidence is not publicly visible.
Institutionally constrained	The layer may be technically possible but faces regulatory, accounting, governance, risk, or incentive friction.
Not applicable yet	The layer is too early-stage or too indirect for the gate to be applied responsibly.

Combined Gate Outcome	Sizing Implication
Retained demand not shown	Not sizing-eligible
Selection not shown	Not sizing-eligible
Institutional friction unresolved	Not yet sizing-eligible, even if technically possible
Retained demand and selection shown, but liquidity burden unclear	Monitoring candidate
Liquidity-density burden visible, but removal sensitivity unproven	Conditional sizing candidate
Removal-sensitive dependence established	Strong sizing candidate
Evidence unavailable but layer structurally close to retained demand	High-priority evidence zone

The five eligibility gates define the evidentiary path from XRP adjacency to filtered liquidity sizing. They ensure that sizing is not triggered by narrative proximity, infrastructure growth, technical availability, or even technical selection alone.

A layer becomes analytically stronger as it moves from selection to retention, from retention to institutional admissibility, from admissibility to liquidity-density burden, and from liquidity burden to removal-sensitive dependence.

5. XRP-Adjacent Institutional Layer Map

The eligibility framework requires a clear definition of the institutional layers to which the gates are applied. This section maps the major XRP-adjacent institutional layers considered in the paper. The purpose of the map is not to identify XRP demand sources in advance. It is to identify the layers where retained-demand, selection, institutional-friction, liquidity-density, and removal-sensitivity evidence should be tested.

The term XRP-adjacent institutional infrastructure refers to institutional products, markets, platforms, or workflows that are meaningfully connected to Ripple, XRPL, RLUSD, XRP liquidity, or XRP-related settlement narratives. The term indicates analytical proximity. It does not imply demand causation.

This distinction is central. A layer may be adjacent to XRP without producing XRP-specific retained demand. It may expand the institutional field without selecting XRP. It may support XRP without requiring it. It may create infrastructure readiness without opening the retained-demand gate. The relevant question for each layer is not whether the layer is associated with XRP, but which eligibility gates, if any, the layer currently approaches or passes.

5.1 Ripple Payments

Ripple Payments is included because payment infrastructure remains one of the most visible contexts in which XRP has historically been discussed. Payment-network expansion may increase corridor coverage, institutional relationships, settlement options, and cross-border workflow relevance. However, payment expansion does not automatically open the retained-demand gate for XRP. The key questions are whether XRP is selected in payment routing, retained by liquidity providers as settlement inventory, and costly to remove from payment execution.

5.2 RLUSD

RLUSD is included because stablecoin infrastructure can alter the need for XRP-specific retained liquidity in multiple directions. It may bypass XRP if RLUSD itself becomes the settlement or treasury asset. It may compress XRP demand if it reduces the need for bridge liquidity. It may create conditional amplification if XRP becomes necessary for RLUSD liquidity routing, cross-ledger movement, market-making inventory, or stress-time liquidity.

5.3 XRPL Base Activity

XRPL base activity includes ledger transactions, account growth, token issuance, fee usage, and general network activity. This layer may show ecosystem development, but it should not be treated as retained-demand evidence by default. It becomes sizing-relevant only if XRPL activity requires institutional XRP retained balances, settlement inventory, collateral, or liquidity-buffer demand.

5.4 XRPL AMM / DEX Infrastructure

XRPL AMM and DEX infrastructure are included because they may directly affect liquidity routing, executable depth, market-making, and slippage conditions. This layer is closer to liquidity-density analysis than general ledger activity, but the mere existence of AMM infrastructure is insufficient. Liquidity must be deep, reliable, institutionally usable, and connected to meaningful workflows.

5.5 Ripple Custody

Ripple Custody is included because custody is often a prerequisite for institutional holding. Custody infrastructure may make XRP holdable, auditable, and operationally admissible. However, custody support does not imply retained demand. It becomes relevant only if custody-held XRP balances are used for collateral, settlement readiness, liquidity buffers, or institutional inventory.

5.6 Ripple Prime / Hidden Road

Ripple Prime / Hidden Road is included because prime brokerage, execution, clearing, financing, margin, and market-making infrastructure may sit close to retained-demand formation. This layer can compress, bypass, or amplify XRP depending on internal design and actual usage. It is one of the most important high-priority evidence zones, even where public evidence remains incomplete.

5.7 Ripple Treasury

Ripple Treasury is included because treasury tooling may alter how institutions manage cash, stablecoins, digital assets, liquidity buffers, settlement balances, and reporting. Treasury infrastructure may reduce idle balances, optimize liquidity, or create new reasons to hold specific assets. It may compress, bypass, or conditionally amplify XRP depending on actual retained balances and institutional admissibility.

5.8 Tokenization / RWA Settlement

Tokenization and real-world asset settlement are included because tokenized markets may generate settlement windows, collateral mobility needs, cross-asset liquidity routing, fragmented market structures, and cross-ledger frictions. These conditions may create a role for external connective liquidity, but they do not automatically select XRP.

5.9 Collateral / Margin Infrastructure

Collateral and margin infrastructure are included because they may produce retained demand without high visible transaction flow. If XRP is actually posted as collateral or margin, retained-demand analysis becomes directly relevant. Eligibility alone does not open the retained-demand gate; actual posting, haircut treatment, concentration limits, and substitution cost matter.

5.10 Derivatives and Hedging

Derivatives and hedging are included because they can alter retained demand through market-making, hedge inventory, margin, collateral, settlement, liquidation, or stress-time liquidity. However, derivatives can also create large notional exposure without requiring retained XRP balances. Notional exposure is not retained demand.

5.11 Market-Making Inventory

Market-making inventory is included because it is one of the clearest retained-demand channels. If market makers or liquidity providers must retain XRP to provide reliable execution, corridor liquidity, AMM depth, cross-venue settlement, or acceptable slippage, XRP-specific retained demand may become measurable.

5.12 Stress-Time Liquidity

Stress-time liquidity is included because normal-state usage may understate the importance of resilience. Under stress, liquidity paths may fragment, spreads may widen, collateral may become less mobile, margin needs may rise, and settlement routes may become less reliable. This layer may be hard to observe, but it is central to the removal-sensitivity gate.

5.13 Cross-Ledger Settlement

Cross-ledger settlement is included because institutional digital finance is unlikely to converge immediately onto one unified ledger. Stablecoins, deposits, tokenized assets, CBDCs, custody systems, and market infrastructures may remain fragmented across platforms. Sizing relevance depends on whether XRP becomes a selected and retained neutral liquidity source across fragmented systems.

5.14 Agent Commerce / Machine Payments

Agent commerce and machine payments are included as an early-stage domain. Automated agents, software systems, devices, or AI-mediated workflows may eventually require programmable settlement, automated routing, low-friction liquidity, or micro-settlement infrastructure. At present, this layer remains low on the evidence gradient unless actual XRP use appears.

The layer map clarifies the scope of the eligibility analysis. XRP-adjacent infrastructure is not treated as a single object. Each layer has different gate relevance. Some layers are closer to retained balances. Others are closer to infrastructure readiness. Some may primarily compress or bypass XRP. Others may become high-priority evidence zones for future sizing eligibility.

6. Evidence-Gated Classification Table

This section applies the evidence gradient and five-gate structure to the XRP-adjacent institutional layers mapped above. The purpose of the classification table is not to produce final empirical conclusions. It is to identify where each layer currently sits on the evidence gradient, how it should be provisionally classified, how confident that classification is, whether the layer is currently eligible for filtered liquidity sizing, and what evidence would change the classification.

The classifications should be read as evidence-gated audit states, not as price signals or investment conclusions.

Layer	Evidence Level	Provisional Classification	Confidence	Sizing Eligibility
Ripple Payments	Level 1-2	Mixed / bypass-leaning	Medium	Not currently eligible
RLUSD	Level 1-2	Bypass-leaning; strategic buffer only if XRP-mediated liquidity appears	Medium	Not currently eligible
XRPL base activity	Level 1	Infrastructure / ecosystem activity, not retained-demand evidence	High	Not eligible
XRPL AMM / DEX	Level 1-2	Conditional liquidity infrastructure	Low-Medium	Conditional, not yet eligible
Ripple Custody	Level 1	Support layer, not necessity	High	Not eligible
Ripple Prime / Hidden Road	Level 1-2 / hard-to-observe	Priority-1 transition layer; conditional sizing candidate if inventory, collateral, or financing evidence appears	Low-Medium	Priority monitor, not yet eligible
Ripple Treasury	Level 1	Compression / bypass-leaning	Medium	Not currently eligible
Tokenization / RWA settlement	Level 1-2	Conditional amplification candidate	Low-Medium	Conditional, not yet eligible
Collateral / margin infrastructure	Level 1 / hard-to-observe	Potential high-density anchor; gate closed until actual posting appears	Low	Closed until actual use evidence
Derivatives and hedging	Level 1-2	Conditional / overread-risk layer	Low-Medium	Not currently eligible
Market-making inventory	Hard-to-observe / potential Level 3	Primary amplification candidate; sizing-ineligible until retained inventory is evidenced	Low	Potential candidate, not yet public-evidence eligible
Stress-time liquidity	Level 0-1 in normal data / event-contingent	Highest-value removal-sensitivity candidate; event-contingent	Low	Conditional only under stress evidence
Cross-ledger settlement	Level 1-2	Conditional amplification candidate	Low-Medium	Conditional, not yet eligible
Agent commerce / machine payments	Level 0-1	Early-stage unresolved domain	Low	Not eligible

Current public evidence does not establish stack-wide XRP-specific amplification. However, the evidence-gated analysis narrows the decisive future evidence zones to a smaller group of layers: market-

making inventory, collateral and margin infrastructure, Ripple Prime / Hidden Road, stress-time liquidity, institutional XRPL liquidity, and cross-ledger settlement.

These layers are not fully sizing-eligible on current public evidence. They are the primary gates through which XRP-specific retained demand could become analytically measurable.

The table should be read in two directions. The defensive direction prevents XRP-adjacent domain expansion from being overread as XRP-specific retained demand. The constructive direction identifies where stronger evidence would matter most. In this sense, the table is not merely a watch list. It is an eligibility map.

Evidence level and structural importance are not the same. A layer may be highly important but low-evidence, as with prime brokerage, market-making inventory, or stress-time liquidity. Conversely, a layer may be highly visible but low in retained-demand significance, as with general XRPL activity, passive custody support, or product announcements. This distinction prevents the analysis from being driven by headlines.

The evidence-gated classification table produces three findings. First, current public evidence supports strong XRP-adjacent domain expansion, but does not establish stack-wide XRP-specific amplification. Second, most visible layers are not currently sizing-eligible. Third, the decisive future evidence zones are narrower than the overall XRP-adjacent ecosystem.

7. Removal-Sensitivity as the Final Gate

The removal-sensitivity gate is the final gate in the sizing-eligibility framework. It asks whether removing XRP from a relevant workflow would materially worsen that workflow relative to available alternatives. This gate is necessary because XRP may be present, supported, selected, or even retained in limited form without becoming structurally meaningful. The strongest claim requires more than presence. It requires consequence.

Paper 4 introduced the removal test as part of the compression-bypass-amplification framework. Paper 5 converts that test into an operational eligibility gate. The question is no longer only whether XRP is part of a stack. The question is whether the stack becomes measurably worse without XRP.

Removal sensitivity does not require monopoly. In institutional financial infrastructure, multiple rails, assets, ledgers, and liquidity sources may coexist. XRP can be structurally meaningful even if it is not the only possible instrument. The relevant standard is whether removing XRP causes measurable degradation under identifiable conditions.

Dimension	Removal-sensitive question
Cost	Does removing XRP increase spreads, fees, funding cost, or execution cost?
Timing	Does removing XRP increase settlement delay or reduce settlement certainty?
Slippage	Does removing XRP worsen execution impact for defined ticket sizes?
Liquidity access	Does removing XRP reduce available depth within acceptable tolerance?
Collateral mobility	Does removing XRP reduce the ability to move or transform collateral efficiently?
Capital efficiency	Does removing XRP increase prefunding, margin, or balance-sheet requirements?
Stress execution	Does removing XRP weaken execution during market disruption?
Operational resilience	Does removing XRP reduce redundancy, fallback capacity, or cross-venue mobility?
Substitution cost	Are substitutes available only at higher cost, lower speed, or weaker resilience?

7.1 Removal Sensitivity and Substitution Cost

Removal sensitivity is inseparable from substitution cost. If XRP is removed, the relevant question is not whether a substitute exists in abstract terms. The relevant question is whether the substitute performs the same function with equal cost, timing, liquidity, capital efficiency, and resilience.

Possible substitutes may include stablecoins, tokenized deposits, fiat rails, internal ledgers, omnibus settlement arrangements, prime-brokerage internalization, CBDC or wholesale settlement rails, bridges or orchestration layers, and other digital assets.

A substitute may be technically available but institutionally inferior under specific conditions. Conversely, a substitute may be institutionally easier to use than XRP. The removal-sensitivity gate therefore compares XRP not to an idealized absence, but to actual available substitutes.

7.2 Removal Impact Dimensions

To avoid treating removal sensitivity as a purely qualitative claim, this paper identifies a set of removal impact dimensions. These dimensions do not form a universal numerical score. They are a conceptual audit checklist for determining whether XRP removal creates measurable degradation relative to actual substitutes.

Removal sensitivity can be evaluated across six observable dimensions: cost, timing, slippage, liquidity access, capital / collateral requirements, and resilience. The purpose is not to imply false precision. The purpose is to clarify that removal sensitivity must eventually connect to observable degradation. A claim that XRP is useful is not enough. The audit question is whether removing XRP worsens one or more of these dimensions relative to actual available substitutes.

7.3 Positive and Negative Removal Triggers

Removal-sensitivity analysis should include both positive and negative triggers. A positive trigger is evidence that removing XRP worsens the workflow. A negative trigger is evidence that the workflow scales without XRP, or that substitute instruments perform the same function without material degradation.

Positive removal triggers include higher cost, longer settlement delay, worse slippage, reduced liquidity access, increased prefunding or collateral requirements, weaker stress-time execution, and reduced resilience or fallback capacity. Negative removal triggers include layer growth while XRP remains unused, substitute instruments performing the function without deterioration, optional XRP support remaining unused, shallow or speculative XRP liquidity despite infrastructure growth, and stress events showing no XRP-specific advantage.

The negative triggers are as important as the positive ones. They allow the framework to identify bypass and compression rather than treating every future development as latent amplification.

7.4 Application Examples

RLUSD

RLUSD can scale as a dollar stablecoin without requiring XRP-mediated liquidity. In that case, RLUSD growth strengthens Ripple-adjacent infrastructure while bypassing XRP. RLUSD becomes more relevant to removal-sensitivity analysis only if XRP-RLUSD liquidity becomes deep and institutionally used, XRP is used to route RLUSD across ledgers or asset classes, or XRP improves RLUSD execution, timing, or resilience relative to stablecoin-only routes.

Ripple Prime / Hidden Road

Ripple Prime / Hidden Road is structurally close to removal sensitivity because prime brokerage can involve execution, financing, margin, collateral, clearing, market-making, and cross-venue liquidity. The question is not whether XRP is tradable or supported, but whether XRP is required for execution quality, financing, collateral mobility, market-making depth, or liquidity access.

Collateral and Margin

Collateral and margin are among the strongest potential retained-demand channels because actual collateral use creates held-form demand. XRP may be eligible as collateral but unused. Eligibility alone does not create retained demand. The layer becomes a high-density anchor only when actual posting, meaningful retained balances, institutional admissibility, and substitution cost are evidenced.

Market-Making Inventory

Market-making inventory is one of the most direct paths to removal sensitivity because market makers may need to retain XRP to support execution depth, corridor liquidity, AMM liquidity, cross-venue routing, or institutional-size settlement. The relevant question is whether XRP inventory improves market quality in a way that substitutes cannot replicate.

Stress-Time Liquidity

Stress-time liquidity may be the strongest but hardest-to-observe removal-sensitivity layer. In normal conditions, a system may appear to function without XRP. During stress, however, liquidity may fragment, spreads may widen, collateral may become less mobile, margin needs may rise, and settlement timing may deteriorate. The question is whether XRP improves execution under disruption.

A strong audit protocol must define not only what confirms a claim, but what weakens it. Evidence that RLUSD scales without XRP-mediated liquidity, prime brokerage grows without XRP inventory or collateral relevance, collateral frameworks expand while XRP remains unused, or stress events occur without XRP-specific resilience would weaken the relevant removal-sensitivity claim.

Removal sensitivity is the final gate because it asks the most important question: not whether XRP is present, but whether its absence matters.

8. High-Priority Evidence Zones

The preceding sections defined the evidence gradient, the five eligibility gates, the institutional friction score, and the removal-sensitivity framework. This section identifies the XRP-adjacent layers where future evidence would matter most for sizing eligibility.

The purpose is not to predict that these layers will produce XRP-specific retained demand. It is to identify where evidence would be most consequential if it appears. A high-priority evidence zone is not a confirmed amplification layer. It is a layer where the evidence, if observed, would be directly relevant to retained demand, institutional admissibility, liquidity-density burden, or removal-sensitive dependence.

The high-priority evidence zones identified in this paper are market-making inventory, collateral and margin infrastructure, Ripple Prime / Hidden Road, stress-time liquidity, XRPL AMM / DEX liquidity, cross-ledger settlement, and RLUSD or tokenization / RWA as conditional secondary zones. These layers are more important than broad ecosystem visibility because they sit closer to the actual gates that determine sizing eligibility.

Priority	Layer	Current Status	Key Positive Trigger	Key Negative Trigger
1	Market-making inventory	Hard-to-observe / primary amplification candidate	LP / MM retained XRP inventory	Liquidity remains speculative or non-institutional
1	Collateral / margin	Potential high-density anchor; gate closed until actual use evidence	Actual XRP posting with usable terms	Eligibility without use; high haircuts
1	Ripple Prime / Hidden Road	Priority-1 transition layer	XRP inventory, collateral, financing, or execution dependence	Prime grows while XRP remains peripheral

2	Stress-time liquidity	Highest-value removal-sensitivity candidate; event-contingent	XRP improves execution under disruption	XRP fails to outperform alternatives under stress
2	XRPL AMM / DEX liquidity	Conditional liquidity infrastructure	Institutional XRP pool usage and depth	Shallow or retail-only liquidity
2	Cross-ledger settlement	Conditional structural candidate	XRP selected as neutral bridge	Stablecoins / orchestration solve without XRP
3	RLUSD	Bypass-leaning unless XRP liquidity appears	XRP-mediated RLUSD liquidity	RLUSD scales independently of XRP
3	Tokenization / RWA	Conditional secondary zone	XRP selected for settlement or collateral mobility	RWA settles via stablecoins / deposits
4	Ripple Custody / XRPL base activity / Agent commerce	Enabling or early-stage	Operational XRP retained balances	Support without retained use

Some developments are visible but weakly connected to retained demand. Product launches, custody support, XRPL activity, stablecoin issuance, tokenization announcements, or broad institutional partnerships may be meaningful at the infrastructure level. However, they do not necessarily show that XRP is retained, selected, institutionally admissible, liquidity-burdensome, or costly to remove.

Other developments may be less visible but more decisive. Market-making inventory, collateral posting, prime-brokerage margin treatment, liquidity-provider balances, stress-time execution, and cross-ledger routing may not appear prominently in public narratives, but they are closer to the evidence gates. The strongest future evidence is likely to come not from broad announcements, but from specific operational signals.

8.1 Proxy Indicators for Hard-to-Observe Layers

Several of the most important evidence zones are difficult to observe directly. Prime-brokerage inventory, market-maker positioning, collateral use, internal routing, liquidity-buffer balances, and stress-time execution behavior may not be publicly disclosed. If the framework required only direct evidence, it would miss important emerging signals. If it accepted weak public narratives as evidence, it would overclaim.

For that reason, hard-to-observe layers should be monitored through proxy indicators. A proxy indicator does not establish sizing eligibility by itself. It identifies a signal that may justify closer scrutiny or movement along the evidence gradient. Proxy evidence should be treated as upgrade-relevant only when it is persistent, workflow-linked, and corroborated by at least one other signal.

Layer	Hard-to-observe evidence	Possible proxy indicators
Ripple Prime / Hidden Road	XRP inventory, client financing, collateral use, internal execution routing	XRP institutional execution depth; OTC liquidity; margin / collateral policy references; client-facing XRP support; financing terms; custody balance changes; market-maker participation; reported volume mix
Market-making inventory	LP / MM retained XRP balances	Order-book depth; spread compression; venue depth stability; AMM depth; liquidity resilience under volatility; reduced slippage for defined ticket sizes; corridor-specific liquidity behavior
Collateral / margin	Actual XRP posting, haircut treatment, concentration limits	Collateral eligibility schedules; margin-policy disclosures; lending-market terms; financing rates; custody-linked collateral products; risk-limit references
Stress-time liquidity	XRP use under market disruption	Spread behavior during stress; relative depth retention; slippage comparison versus substitutes; settlement reliability; liquidity-provider behavior; cross-venue mobility

RLUSD liquidity routing	XRP-mediated RLUSD liquidity	XRP-RLUSD pool depth; routing references; cross-ledger liquidity behavior; market-maker quotes; institutional pair usage; stress-time liquidity performance
XRPL AMM / DEX	Institutional use of XRP pools	Pool depth; trade-size capacity; slippage at defined sizes; professional LP participation; integration with RLUSD / RWA workflows; liquidity stability

8.2 Exit and Bypass Criteria

An operational audit protocol must define not only what opens a gate, but what closes it. Without exit or bypass criteria, every unresolved layer risks becoming a permanent source of speculative optionality. That would undermine the discipline of the framework.

These criteria are intentionally strict. They prevent the framework from preserving every layer as an indefinite future option. A layer that repeatedly fails to produce selection, retained demand, institutional admissibility, liquidity burden, or removal sensitivity should not remain indefinitely open.

Exit criteria do not require immediate abandonment after a single negative signal. Institutional infrastructure develops slowly, and public evidence may be incomplete. However, persistent evidence of bypass or compression should update the classification. This is not pessimism. It is input discipline.

The positive and negative trigger conditions summarized in this section can be operationalized through a trigger matrix. For readability, the full matrix is provided in Appendix A as part of the Operational Audit Checklist.

High-priority evidence zones convert the eligibility framework into a practical monitoring protocol. They show where future evidence would matter most and where visible ecosystem growth should not be overread.

The relevant issue is no longer whether XRP is adjacent to institutional infrastructure. The relevant issue is whether a limited set of inventory, collateral, liquidity, and stress-time layers can pass the eligibility gates. That narrowing is the practical contribution of Paper 5.

9. Relationship to Filtered Sizing and Public Sizing Literature

The preceding sections defined the evidence gradient, the five eligibility gates, institutional friction scoring, removal-sensitivity analysis, proxy indicators, exit criteria, and high-priority evidence zones. This section clarifies how that framework relates to downstream filtered liquidity sizing and to public XRP sizing literature.

The key distinction is simple: sizing is not rejected. Sizing is gated.

Filtered liquidity sizing can be analytically useful when applied to layers that have passed the relevant evidence gates. It becomes dangerous when applied to layers that are merely adjacent, visible, or narratively attractive. The purpose of this paper is therefore not to prevent sizing, but to define the evidentiary conditions under which sizing becomes legitimate.

Liquidity sizing is input-sensitive. Its outputs depend heavily on the assumptions fed into it: flow size, usable float, turnover, executable depth, slippage tolerance, collateral lockup, market-making inventory, stress conditions, and substitution cost. If those inputs are poorly grounded, the output may appear precise while resting on unsupported assumptions.

The audit protocol in this paper is therefore an input-quality control mechanism. It asks whether a layer has produced enough evidence to be admitted into filtered sizing. A layer that fails the gates should not be sized merely because it is large or institutionally important. A layer that passes the gates may become appropriate for conditional liquidity-density analysis.

Public sizing literature	Paper 5
How large must the pipe be?	Is there enough evidence that the pipe is required?
Scenario sizing	Evidence-gated eligibility
Liquidity burden	Gate-tested retained demand
Price-sensitive outputs	Non-price operational audit protocol
Conditional flow assumptions	Input-quality control
Sizing model	Admissibility test

This distinction is complementary rather than competitive. A public sizing model may be useful once a layer has passed the gates. For example, if evidence showed that market makers retain XRP inventory to support institutional execution, filtered sizing could examine required inventory, turnover, slippage tolerance, and executable depth. If XRP were actually posted as collateral or margin, filtered sizing could examine locked balances, haircut effects, substitution cost, and liquidity-density burden. If stress-time liquidity evidence emerged, filtered sizing could model required depth under disruption.

But without those evidence gates, the sizing model lacks admissible input. This paper therefore functions as a pre-sizing admissibility layer.

Filtered sizing should not claim what XRP should be worth in general, a universal price target, asset indispensability across the entire stack, that all XRP-adjacent flows require XRP, that ecosystem growth automatically becomes retained demand, or that conditional scenarios are forecasts.

A model that sizes every adjacent narrative becomes noisy. A model that sizes only gate-passing layers becomes sharper.

Only then is sizing earned.

10. Conclusion

This paper has developed an operational audit protocol for determining when XRP-adjacent institutional infrastructure becomes eligible for downstream filtered liquidity sizing. It does not provide a price forecast, investment thesis, or claim of XRP indispensability. It asks a prior question: what evidence must be present before an XRP-adjacent layer can responsibly be admitted into a sizing exercise?

The preceding papers established the audit discipline. Paper 1 distinguished visible usage from retained demand. Paper 2 distinguished infrastructure expansion from settlement closure. Paper 3 distinguished conditional liquidity sizing from price narrative. Paper 4 distinguished stack expansion from asset-specific amplification through the categories of compression, bypass, and amplification. The present paper extends that sequence by defining the evidentiary and institutional conditions under which a layer may move from stack audit into filtered sizing.

The central contribution is not another price model. It is an admissibility protocol. Before a layer can be sized, it must pass through a disciplined evidentiary sequence: retained demand, selection, institutional friction, liquidity-density burden, and removal sensitivity. These gates are designed to prevent narrative proximity, infrastructure growth, or technical compatibility from being converted prematurely into quantitative assumptions.

The paper's evidence gradient provides a way to classify incomplete public evidence without either overstating or discarding it. A layer may be adjacent but unselected. It may be infrastructure-ready but institutionally constrained. It may show selection but no retained balances. It may show retained demand without meaningful liquidity-density burden. It may show liquidity burden without removal-sensitive dependence. These distinctions allow the analysis to remain both cautious and operational.

The paper's main substantive conclusion is limited but meaningful. Current public evidence supports strong XRP-adjacent domain expansion, but it does not establish stack-wide XRP-specific amplification.

Ripple-related, RLUSD-related, XRPL-related, custody-related, treasury-related, tokenization-related, and prime-brokerage-related developments should not be collapsed into a single asset-demand claim. Adjacency is not demand. Technical support is not selection. Selection is not retained demand. Retained demand is not sizing eligibility. And sizing eligibility is not established unless institutional admissibility and removal sensitivity can be shown.

At the same time, the analysis does not end in generalized uncertainty. The sizing-eligibility framework narrows the decisive evidence zones to a smaller set of layers. The most important future evidence is likely to appear not in broad ecosystem headlines, but in specific operational channels: market-making inventory, collateral and margin infrastructure, Ripple Prime / Hidden Road, stress-time liquidity, institutional XRPL liquidity, and cross-ledger settlement.

The institutional friction gate is especially important. Technical feasibility alone is not institutional adoption. XRP may be technically efficient, liquid, or settlement-capable, yet still fail to become sizing-relevant if institutions cannot approve, hold, account for, risk-manage, collateralize, or govern it within real operating constraints. A serious XRP retained-demand thesis must therefore pass not only market and liquidity tests, but also institutional admissibility tests.

The removal-sensitivity gate is the final test. A layer becomes structurally meaningful not merely when XRP is present, but when XRP's absence matters. If XRP can be removed without material degradation in cost, timing, slippage, liquidity access, collateral mobility, capital efficiency, stress-time execution, or operational resilience, then the stronger sizing claim remains weak. If removal produces measurable deterioration under identifiable conditions, then XRP-specific retained demand becomes far more analytically consequential.

The protocol is intentionally bilateral. It defines not only what would open a gate, but also what would keep a gate closed. If RLUSD scales while XRP-mediated liquidity remains absent, the RLUSD layer should remain bypass-leaning for XRP-specific sizing purposes. If prime-brokerage infrastructure expands while XRP remains peripheral to inventory, collateral, financing, and execution workflows, the prime layer should remain outside filtered sizing. If collateral markets grow while XRP remains unposted or unattractive under haircut and risk-limit conditions, the collateral gate remains closed. If stress events occur and XRP does not improve depth, timing, slippage, or resilience relative to substitutes, the removal-sensitivity claim weakens.

This is the reason the framework is an audit protocol rather than a narrative screen. It can open gates, but it can also close them. It can identify high-priority evidence zones, but it can also exclude layers that remain adjacent, optional, institutionally constrained, liquidity-irrelevant, or removable without degradation.

Paper 5 therefore marks a transition in the series. Papers 1 through 4 built the audit architecture. Paper 5 turns that architecture into an operational eligibility protocol. It asks not only whether XRP-adjacent infrastructure is growing, but whether the evidence has matured enough to justify quantitative treatment.

The conclusion is therefore neither promotional nor dismissive. XRP-specific amplification is not established across the stack on current public evidence. But the universe of relevant evidence is now narrower and more operationally defined. The decisive question is whether a limited set of layers - market-making inventory, collateral and margin, prime brokerage, stress-time liquidity, institutional XRPL liquidity, and cross-ledger settlement - can produce evidence of retained demand, institutional admissibility, liquidity-density burden, and removal-sensitive dependence.

In the sequence of the series, Paper 5 fixes the boundary between evidence and sizing. The earlier papers establish what retained demand, closure competition, liquidity-density burden, and stack outcomes mean; this paper determines when those concepts are sufficiently evidenced to become admissible sizing inputs.

Until then, filtered sizing should remain gated. If those gates open, sizing becomes earned.

Appendix A: Operational Audit Checklist

This appendix provides a practical checklist for applying the operational audit protocol developed in this paper. It is not a separate valuation model, price framework, or investment tool. Its purpose is to help classify XRP-adjacent institutional layers according to evidence maturity, eligibility gates, institutional friction, proxy indicators, trigger conditions, and sizing eligibility status.

The checklist should be applied conservatively. A layer should not be elevated merely because it is visible, strategically important, or associated with Ripple, XRPL, RLUSD, or XRP narratives. Elevation requires evidence that is workflow-linked, institutionally meaningful, and relevant to retained demand, liquidity-density burden, or removal-sensitive dependence.

A.1 Evidence Level

Evidence Level	Evidence State	Audit Question
Level 0	Narrative adjacency only	Is the layer merely associated with XRP, Ripple, XRPL, RLUSD, or institutional digital-asset narratives?
Level 1	Infrastructure ready / selection pending	Does infrastructure exist that could support XRP use, without evidence of actual selection?
Level 2	Selection indicated / institutional admissibility unproven	Is XRP selected, usable, or integrated, but without evidence of institutional retention or admissibility?
Level 3	Retained demand / institutional use evidence emerging	Is there workflow-linked holding behavior showing XRP retained in operational form?
Level 4	Liquidity-density burden visible	Does XRP-specific retained demand create meaningful burden on float, depth, turnover, positioning, or slippage?
Level 5	Removal-sensitive dependence established	Would removing XRP materially worsen cost, timing, slippage, liquidity access, capital efficiency, or resilience?

A.2 Eligibility Gates

Gate	Audit Question	Status
Retained Demand Gate	Is XRP held, positioned, posted, buffered, or inventoried in operational form?	Closed / Partial / Open
Selection Gate	Is XRP actually selected within the relevant workflow?	Closed / Partial / Open
Institutional Friction Gate	Can XRP be approved, retained, governed, risk-managed, and operationally relied upon?	F0-F5
Liquidity-Density Burden Gate	Does XRP-specific retained demand create meaningful burden on float, depth, turnover, positioning, or slippage?	Closed / Partial / Open
Removal-Sensitivity Gate	Would removing XRP materially worsen the workflow relative to actual substitutes?	Closed / Partial / Open

A.3 Institutional Friction Score

Score	Institutional state
F0	Not institutionally admissible
F1	Technically supported, institutionally unresolved
F2	Operationally available, balance-sheet constrained
F3	Institutionally usable under limited conditions
F4	Broadly admissible with manageable constraints
F5	Fully integrated institutional workflow asset

A.4 Proxy Evidence Standard

Criterion	Meaning
Persistent	The signal appears repeatedly or over time, rather than as a one-off event.

Workflow-linked	The signal connects to a real operational process, not merely general market activity.
Corroborated	The signal is supported by at least one additional indicator.

A.5 Positive and Negative Trigger Matrix

Layer	Positive Trigger	Negative Trigger	Gate Affected
RLUSD	XRP-mediated RLUSD liquidity becomes institutionally used	RLUSD scales independently of XRP	Selection / Removal-Sensitivity
Ripple Payments	XRP-retained liquidity provider inventory or XRP-mediated settlement dependence appears	Payment flows expand through fiat, stablecoins, internal liquidity, or prefunding optimization without XRP-retained inventory	Retained Demand / Removal-Sensitivity
Ripple Prime / Hidden Road	XRP inventory, collateral, financing, or execution dependence appears	Prime grows while XRP remains peripheral	Retained Demand / Institutional Friction
Market-making inventory	LP / MM retained XRP inventory supports execution or corridors	Liquidity remains speculative or non-institutional	Retained Demand / Liquidity-Density
Collateral / margin	Actual XRP posting with usable haircut and risk terms	Eligibility exists but XRP remains unposted or unattractive	Retained Demand / Institutional Friction
XRPL AMM / DEX	Institutional XRP pool usage with meaningful depth	Pools remain shallow, retail, or disconnected from workflows	Liquidity-Density
Cross-ledger settlement	XRP selected as neutral bridge with retained balances	Stablecoins / orchestration solve without XRP	Selection / Removal-Sensitivity
Stress-time liquidity	XRP improves execution under disruption	XRP underperforms alternatives during stress	Removal-Sensitivity
Tokenization / RWA	XRP selected for settlement or collateral mobility	RWA settles via stablecoins / deposits	Selection
Ripple Treasury	XRP retained as buffer or settlement inventory	Treasury optimization bypasses XRP	Retained Demand / Compression
Ripple Custody	Custody-held XRP used operationally	Custody support remains passive	Retained Demand
Agent commerce	XRP used in automated payment routing with retained balances	Automated payments scale through stablecoins, bank APIs, or platform balances without XRP	Selection / Retained Demand

A.6 Exit and Bypass Criteria

Layer	Exit / Bypass Criteria	Implication
RLUSD	RLUSD scales materially while XRP-RLUSD liquidity remains shallow, unused, or non-institutional	RLUSD should remain bypass-leaning rather than amplification-relevant
Ripple Payments	Payment flows expand through fiat, stablecoins, internal liquidity, or prefunding optimization without XRP-retained inventory	Payment expansion should not be treated as XRP retained demand
Ripple Prime / Hidden Road	Prime services expand while XRP remains trading-only and absent from collateral, financing, inventory, or execution-dependence workflows	Prime layer remains high-priority but not sizing-eligible
Ripple Treasury	Treasury adoption grows while XRP is not retained as buffer, settlement asset, or liquidity inventory	Treasury layer should be treated as compression / bypass-leaning
XRPL AMM / DEX	XRP pool growth remains shallow, speculative, retail-oriented, or disconnected from institutional workflows	AMM / DEX remains infrastructure-ready but not sizing-eligible
Collateral / margin	Collateral markets expand while XRP remains unposted, unattractive, highly haircut, or concentration-limited	Collateral gate remains closed
Market-making inventory	Market activity grows without evidence of professional retained XRP inventory	Market activity should not be treated as retained demand

Cross-ledger settlement	Cross-ledger workflows are solved by stablecoins, tokenized deposits, bridges, orchestration, or internal ledgers without XRP selection	Cross-ledger layer remains bypass / non-sizing-eligible
Tokenization / RWA	Tokenized assets settle through stablecoins, deposits, internal ledgers, or platform-specific systems without XRP	RWA expansion should not be treated as XRP demand
Stress-time liquidity	Stress events occur and XRP does not improve depth, slippage, timing, or resilience relative to substitutes	Removal-sensitivity claim weakens
Agent commerce	Automated payment systems scale through stablecoins, bank APIs, or platform balances without XRP	Agent-commerce layer remains non-sizing-eligible

A.7 Sizing Eligibility Decision

Decision	Meaning
Not eligible	Required gates are closed.
Deferred	Evidence is incomplete, but the layer remains structurally relevant.
Priority monitor	The layer is close to retained-demand or removal-sensitivity formation.
Conditional candidate	Some gates are partially open, but stronger evidence is required.
Eligible for filtered sizing	Retained demand, selection, institutional admissibility, liquidity burden, and removal sensitivity are sufficiently evidenced.

A.8 Compact Audit Template

Audit Item	Result
Layer	-
Evidence Level	Level 0 / 1 / 2 / 3 / 4 / 5
Retained Demand Gate	Closed / Partial / Open
Selection Gate	Closed / Partial / Open
Institutional Friction Score	F0 / F1 / F2 / F3 / F4 / F5
Liquidity-Density Burden Gate	Closed / Partial / Open
Removal-Sensitivity Gate	Closed / Partial / Open
Positive Trigger	-
Negative Trigger	-
Proxy Evidence	None / Weak / Moderate / Strong
Sizing Eligibility	Not eligible / Deferred / Priority monitor / Conditional candidate / Eligible
Classification	Compression / Bypass / Amplification / Mixed / Unresolved
Update Condition	-

This compact template summarizes the operational function of the paper: to decide whether a layer should remain outside filtered sizing, be monitored as a high-priority evidence zone, or be elevated toward conditional sizing eligibility.

Footnotes

1. This paper is the fifth paper in a series on XRP-adjacent institutional infrastructure, retained demand, and liquidity-density analysis. It should be read as an applied working paper rather than a price forecast, valuation exercise, or investment thesis.

2. The term 'XRP-adjacent institutional infrastructure' is used analytically. It indicates proximity to XRP, Ripple, XRPL, RLUSD, or related institutional settlement narratives. It does not imply asset-specific retained demand.

3. Public sizing literature is treated as useful but downstream. The present paper does not reject sizing; it defines the evidence standard required before sizing inputs become admissible.

4. All classifications are provisional and evidence-gated. They are based on public evidence and should not be interpreted as investment advice, price targets, or claims about future market performance.

Keywords

Evidence Before Sizing; XRP; Retained Demand; Filtered Liquidity Sizing; Institutional Friction; Removal Sensitivity; Liquidity Density; Market-Making Inventory; Collateral and Margin; Stress-Time Liquidity; Cross-Ledger Settlement; Ripple Prime; Hidden Road; RLUSD; XRPL; Operational Audit; Financial Market Infrastructure

Author Note

This paper is presented as a working paper and should be read as a framework-oriented contribution. It is the fifth paper in a retained-demand audit series for institutionally connected digital assets, translating the retained-demand, closure, liquidity-density, and stack-outcome concepts developed in the preceding papers into an operational eligibility protocol.

The purpose of the paper is not to argue that XRP is indispensable, nor to deny that XRP may become relevant under certain institutional conditions. Its purpose is to define the evidentiary and institutional conditions under which an XRP-adjacent layer may become eligible for downstream filtered liquidity sizing.

XRP-adjacent infrastructure is not treated as XRP retained demand. Adjacency, technical support, product integration, or ecosystem expansion may justify monitoring, but they do not justify sizing unless retained demand, selection, institutional admissibility, liquidity-density burden, and removal sensitivity are sufficiently evidenced.

All classifications in the paper are provisional and evidence-gated. They are based on public evidence and should not be interpreted as investment advice, price targets, recommendations to buy, sell, or hold any digital asset, or claims about future market performance.

Short Description

This paper provides an evidence-gated operational audit protocol for determining when XRP-adjacent institutional infrastructure becomes eligible for downstream filtered liquidity sizing. It separates XRP adjacency from XRP retained demand and defines the evidence standards, institutional constraints, removal-sensitivity tests, proxy indicators, and exit criteria required before XRP-adjacent narratives can be admitted into liquidity-density analysis.

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